Nathan Tran

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scholar.google.com/citations?user=fZ4rutkAAAAJ

EDUCATION

North Carolina State University Raleigh, NC
B.S. Computer Science, Artificial Intelligence & Machine Learning Specialization Expected May 2027
Graduate plans: M.S./Ph.D. in Computer Science, beginning Fall 2027

- Honors: Dean's List (Fall 2023, Spring 2025, Fall 2025); College of Engineering Research Experience for Undergraduates (REU), Summer 2026
- Coursework: Data Structures & Algorithms (CSC 316), Software Development Fundamentals (CSC 216), Machine Learning (CSC 422), Artificial Intelligence (CSC 411), Independent Research (CSC 499)

EXPERIENCE

Undergraduate Researcher, Extended Horizon Lab, NC State University Feb 2026 – Present

- Integrated a ZED 2i stereo camera into Playground+, a projection-based mixed-reality system, using CUDA on an RTX 4060 while preserving webcam compatibility; raised real-time performance from ~15 FPS at 480p to 60 FPS at 720p.
- Independently built a gaze-aware VR research system (Meta Quest 3, Pupil Labs Neon XR, Unity/C#) that uses gaze, fixation, pupil, and dwell-time signals to trigger proactive AI-agent explanations.
- Planned XR deployment hardware (cameras, projectors, compute, physical setup, budget) and led part of a Ph.D. student's project as the lab's only undergraduate with an independent project.
- Contributed to UIST and CHI research: prototyping, user studies, qualitative analysis, research writing, and a CHI submission video.

RESEARCH & PUBLICATIONS

- PulseCoach: state-aware AI for AR using physiological and motion sensing (EDA, HRV, IMU). ACM UIST 2026 Student Innovation Contest submission.
- MRPilot: mixed-reality user study (recruitment, interviews, qualitative analysis, demo video). Under review at CHI 2027; preprint at arxiv.org/abs/2610.07477.

PROJECTS

Job Application Organizer | Java, React

- Full-stack job tracker with Java backend logic and React UI; State design pattern for status transitions, plus search, filter, sort, and simulated REST updates.

Facial Recognition ATM | C++, Python, OpenCV, SQLite

- Facial-authentication ATM using Haar cascades and LBPH, with account balances, withdrawals, transaction logging, and separated auth/UI/database layers.

Used-Car Price Prediction | Python, CatBoost, SHAP

- ML model on a 4,009-vehicle dataset with engineered features (vehicle age, miles per year) and SHAP explainability; diagnosed overfitting (train R^2 0.96 vs. test 0.85) and iterating on validation.

SKILLS

Languages: Python, Java, C, C++, C#, JavaScript, SQL
Frameworks & Tools: Unity, React, Spring Boot, OpenCV, CUDA, Git, Linux, JUnit, GDB, SQLite
AI/XR: Computer Vision, Eye Tracking, AI Agents, Multimodal Sensing, Machine Learning (CatBoost, SHAP)

LEADERSHIP

- Head Coach, Highland Creek Hurricanes Swim Team: manage 4-5 coaches and 100+ swimmers, plan practices and meets.
- President, NC State Olympic Weightlifting Club: budgeting, recruiting, sponsorships, and coaching while competing.